

SOC/L1 - 6.2. Wireshark The Basics

Task 1 - Introduction

Wireshark is an open-source, cross-platform network packet analyser tool capable of sniffing and investigating live traffic and inspecting packet captures (PCAP). It is commonly used as one of the best packet analysis tools.

Answer the questions below

- | Which file is used to simulate the screenshots? http1.pcapng
- | Which file is used to answer the questions? Exercise.pcapng

Task 2 - Tool Overview

Use Cases: Wireshark is one of the most potent traffic analyser tools available in the wild. There are multiple purposes for its use:

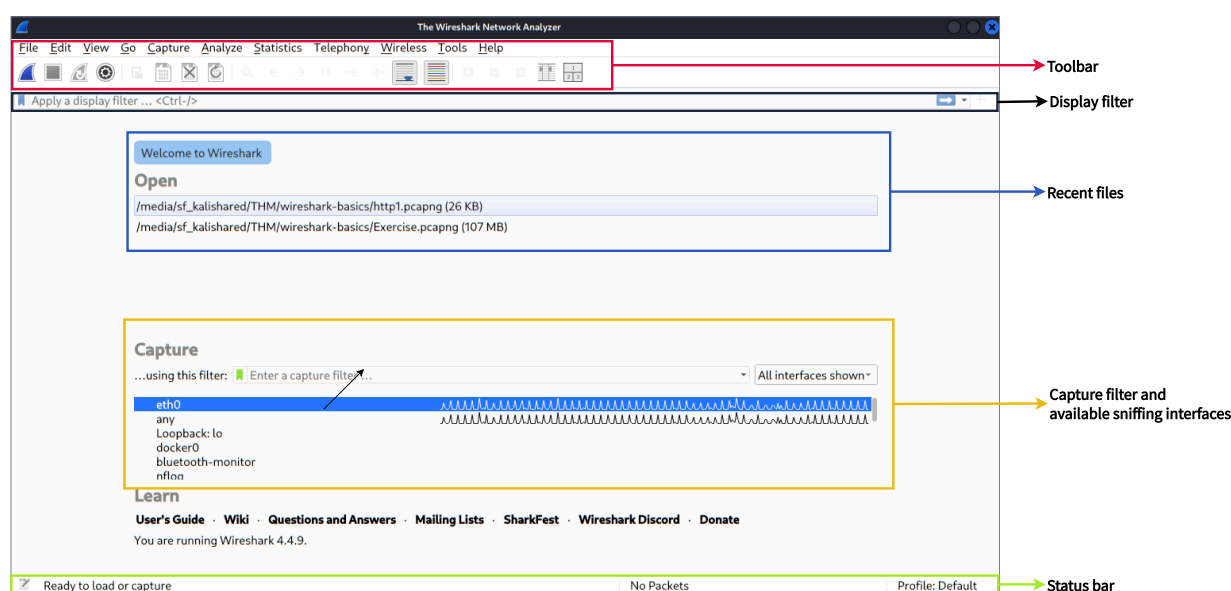
- Detecting and troubleshooting network problems, such as network load failure points and congestion.
- Detecting security anomalies, such as rogue hosts, abnormal port usage, and suspicious traffic.
- Investigating and learning protocol details, such as response codes and payload data.

Note: Wireshark is not an Intrusion Detection System (IDS). It only allows analysts to discover and investigate the packets in depth. It also doesn't modify packets; it reads them. Hence, detecting any anomaly or network problem highly relies on the analyst's knowledge and investigation skills.

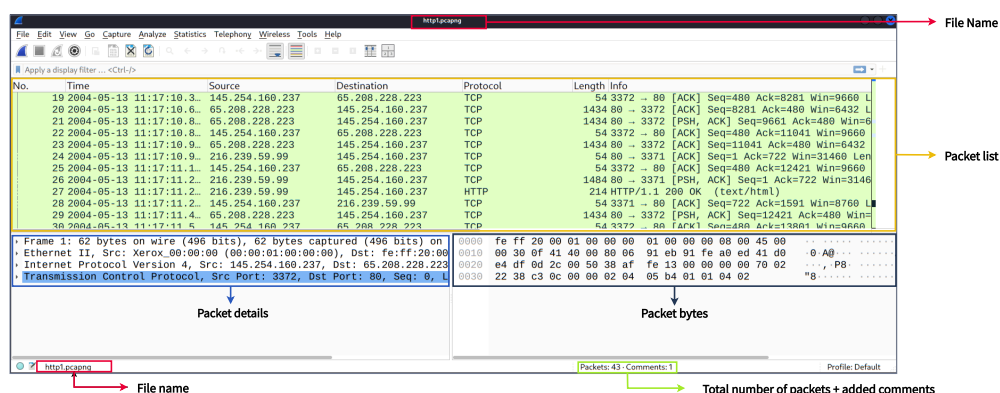
GUI and Data

Toolbar	The main toolbar contains multiple menus and shortcuts for packet sniffing and processing, including filtering, sorting, summarising, exporting and merging.
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Display Filter Bar	The main query and filtering section.
Recent Files	List of the recently investigated files. You can recall listed files with a double-click.
Capture Filter and Interfaces	Capture filters and available sniffing points (network interfaces). The network interface is the connection point between a computer and a network. The software connection (e.g., lo, eth0 and ens33) enables networking hardware.
Status Bar	Tool status, profile and numeric packet information.

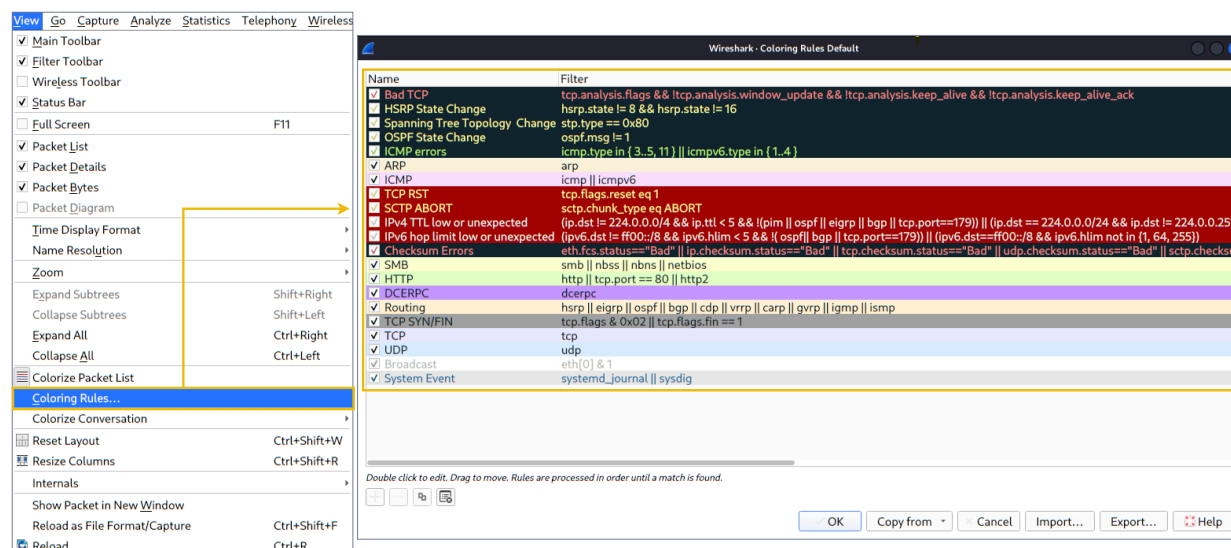


Loading PCAP Files: Dragging and dropping the file, or double-clicking on the file to load a pcap.

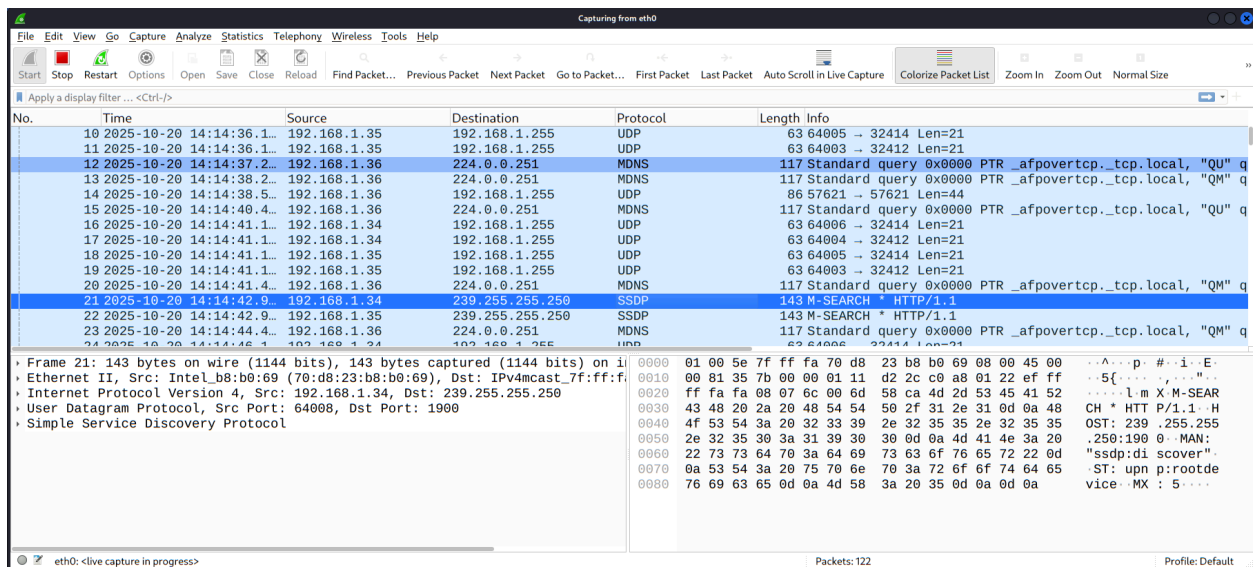


Packet List Pane	Summary of each packet (source and destination addresses, protocol, and packet info). You can click on the list to choose a packet for further investigation. Once you select a packet, the details will appear in the other panels.
Packet Details Panel	Detailed protocol breakdown of the selected packet.
Packet Bytes Pane	Hex and decoded ASCII representation of the selected packet. It highlights the packet field depending on the clicked section in the details pane.

Colouring Packets: "View → Coloring Rules" menu to create permanent colouring rules.



Traffic Sniffing: The blue "**shark button**" to start network sniffing (capturing traffic), the red button will stop the sniffing, and the green button will restart the sniffing process.

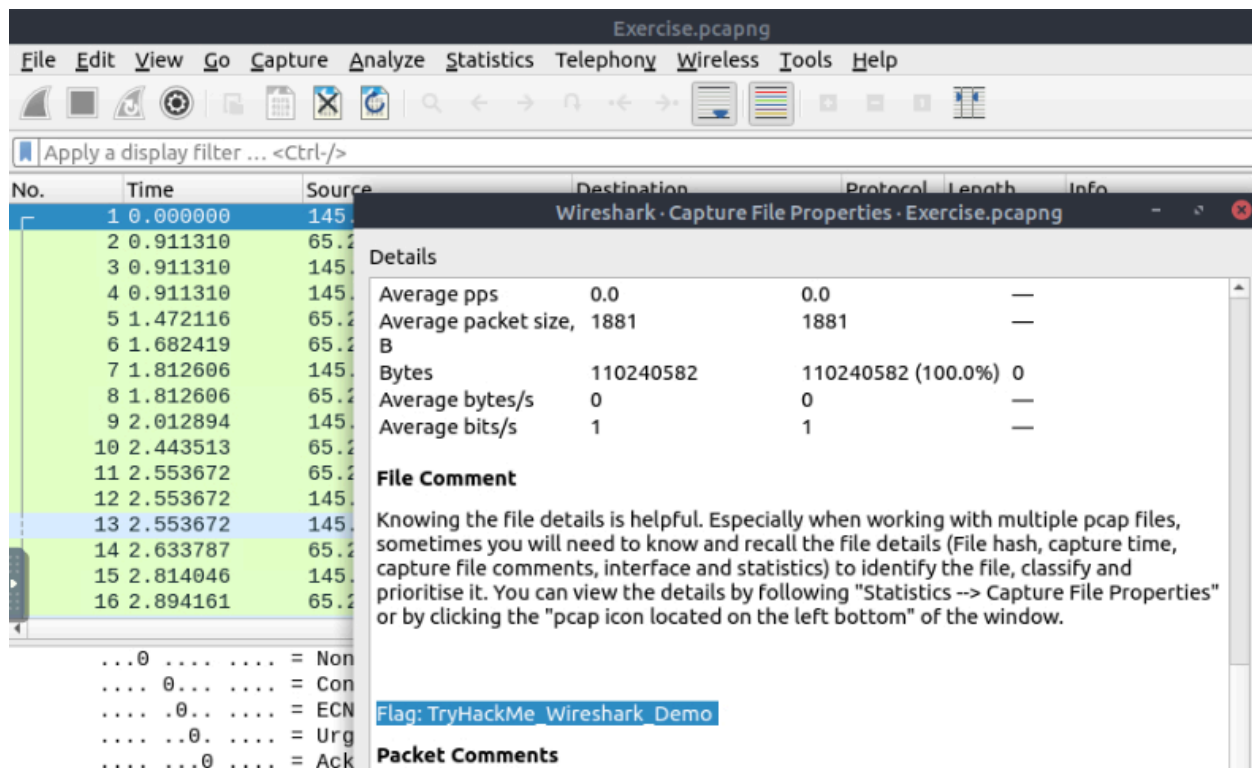


Merge PCAP Files: You can use the **"File → Merge"** menu path to merge a pcap with the processed one.

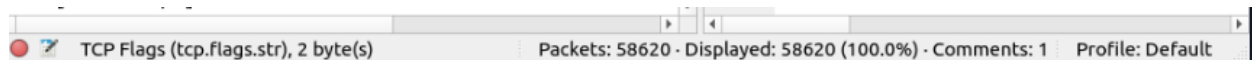
View File Details: You can view the details by following **"Statistics → Capture File Properties"** or by clicking the **"pcap icon located on the left bottom"**.

Use the "Exercise.pcapng" file to answer the questions. Read the "capture file comments". What is the flag?

TryHackMe_Wireshark_Demo

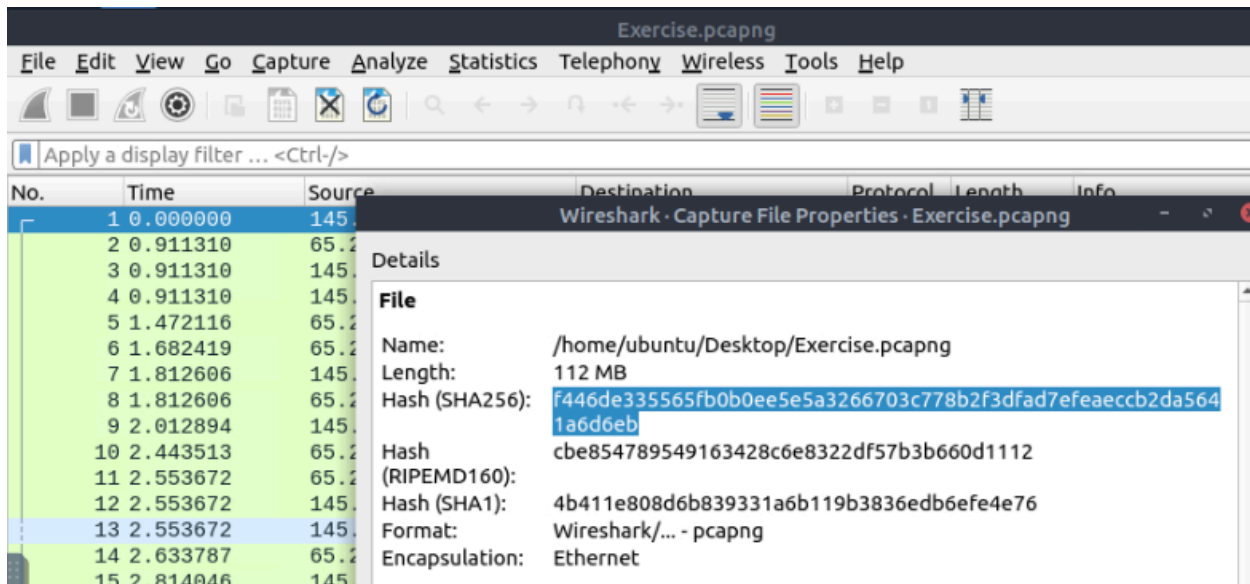


What is the total number of packets? 58620



What is the SHA256 hash value of the capture file?

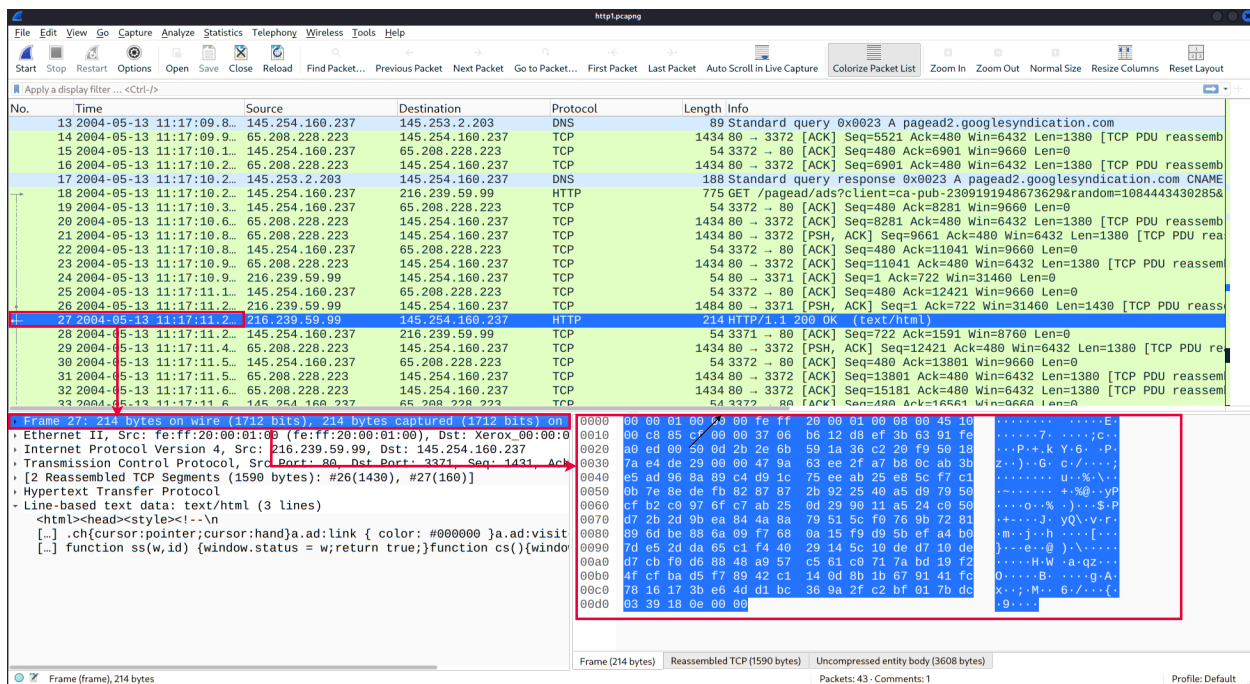
f446de335565fb0b0ee5e5a3266703c778b2f3dfad7efeaeccb2da5641a6d6eb



Task 3 - Packet Dissection

Wireshark supports a long list of protocols for dissection, and you can also write your dissection scripts.

Packet Details: You can click on a packet in the packet list pane to open its details (double-click will open details in a new window). Packets consist of 5 to 7 layers based on the OSI model.



Each time you click a detail, it will highlight the corresponding part in the packet bytes pane.

Frame 27: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface
 Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00), Dst: Xerox_00:00:00 (00:00:00:00:00:00)
 Internet Protocol Version 4, Src: 216.239.59.99, Dst: 145.254.160.237
 Transmission Control Protocol, Src Port: 80, Dst Port: 3371, Seq: 1431, Ack: 722, Len: 1590
 [2 Reassembled TCP Segments (1590 bytes): #26(1430), #27(160)]
 Hypertext Transfer Protocol
 Line-based text data: text/html (3 lines)
 <html><head><style><!--

 [...] .ch{cursor:pointer;cursor:hand}a.ad:link { color: #000000 }a.ad:visited { color
 [...] function ss(w,id) {window.status = w;return true;}function cs(){window.status=''

Frame (214 bytes) | Reassembled TCP (1590 bytes) | Uncompressed entity body (3608 bytes)

- ▶ Frame 27: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on in
- ▶ Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00), Dst: Xerox_00:00:00
- ▶ Internet Protocol Version 4, Src: 216.239.59.99, Dst: 145.254.160.237
- ▶ Transmission Control Protocol, Src Port: 80, Dst Port: 3371, Seq: 1431, Ack:
- ▶ [2 Reassembled TCP Segments (1590 bytes): #26(1430), #27(160)]
- ▶ Hypertext Transfer Protocol
- ▶ Line-based text data: text/html (3 lines)

We can see seven distinct layers to the packet: **frame/packet**, **source [MAC]**, **source [IP]**, **protocol**, **protocol errors**, **application protocol**, and **application data**. Below we will go over the layers in more detail.

The Frame (Layer 1): This will show you what frame/packet you are looking at and details specific to the Physical layer of the OSI model.

- ▶ Frame 27: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits)
 - Section number: 1
 - ▶ Interface id: 0 (unknown)
 - Encapsulation type: Ethernet (1)
 - Arrival Time: May 13, 2004 11:17:11.266912000 BST
 - UTC Arrival Time: May 13, 2004 10:17:11.266912000 UTC
 - Epoch Arrival Time: 1084443431.266912000
 - [Time shift for this packet: 0.000000000 seconds]
 - [Time delta from previous captured frame: 0.040058000 seconds]
 - [Time delta from previous displayed frame: 0.040058000 seconds]
 - [Time since reference or first frame: 3.955688000 seconds]
 - Frame Number: 27
 - Frame Length: 214 bytes (1712 bits)
 - Capture Length: 214 bytes (1712 bits)
 - [Frame is marked: False]
 - [Frame is ignored: False]
 - [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
 - [Coloring Rule Name: HTTP]
 - [Coloring Rule String: http || tcp.port == 80 || http2]

Source [MAC] (Layer 2):This will show you the source and destination MAC Addresses; from the Data Link layer of the OSI model.

```
↳ Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00), Dst: Xerox_00:00:00 (00:00:01:00:00:00)
  ↳ Destination: Xerox_00:00:00 (00:00:01:00:00:00)
  ↳ Source: fe:ff:20:00:01:00 (fe:ff:20:00:01:00)
    Type: IPv4 (0x0800)
    [Stream index: 0]
```

Source [IP] (Layer 3):This will show you the source and destination IPv4 Addresses; from the Network layer of the OSI model.

```
↳ Internet Protocol Version 4, Src: 216.239.59.99, Dst: 145.254.160.237
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ↳ Differentiated Services Field: 0x10 (DSCP: Unknown, ECN: Not-ECT)
    Total Length: 200
    Identification: 0x85cf (34255)
  ↳ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 55
    Protocol: TCP (6)
    Header Checksum: 0xb612 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 216.239.59.99
    Destination Address: 145.254.160.237
    [Stream index: 2]
```

Protocol (Layer 4):This will show you details of the protocol used (UDP/TCP) and source and destination ports; from the Transport layer of the OSI model.

- Transmission Control Protocol, Src Port: 80, Dst Port: 3371, Seq: 1431, Ack: 722, Len: 160
 - Source Port: 80
 - Destination Port: 3371
 - [Stream index: 1]
 - [Stream Packet Number: 4]
 - [Conversation completeness: Incomplete (12)]
 - [TCP Segment Len: 160]
 - Sequence Number: 1431 (relative sequence number)
 - Sequence Number (raw): 778787098
 - [Next Sequence Number: 1591 (relative sequence number)]
 - Acknowledgment Number: 722 (relative ack number)
 - Acknowledgment number (raw): 918692089
 - 0101 = Header Length: 20 bytes (5)
 - Flags: 0x018 (PSH, ACK)
 - Window: 31460
 - [Calculated window size: 31460]
 - [Window size scaling factor: -1 (unknown)]
 - Checksum: 0xde29 [unverified]
 - [Checksum Status: Unverified]
 - Urgent Pointer: 0
 - [Timestamps]
 - [SEQ/ACK analysis]
 - TCP payload (160 bytes)
 - TCP segment data (160 bytes)

Protocol Errors: This continuation of the 4th layer shows specific segments from TCP that needed to be reassembled.

- [2 Reassembled TCP Segments (1590 bytes): #26(1430), #27(160)]
 - [Frame: 26, payload: 0-1429 (1430 bytes)]
 - [Frame: 27, payload: 1430-1589 (160 bytes)]
 - [Segment count: 2]
 - [Reassembled TCP length: 1590]
 - [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a5033503a20706f6c6963797265663d22687474703a2f2...

Application Protocol (Layer 5): This will show details specific to the protocol used, such as HTTP, FTP, and SMB. From the Application layer of the OSI model.

- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - P3P: policyref="http://www.googleadservices.com/pagead/p3p.xml", CP="NOI DEV PSA PSD IVA PVD OTP OUR OTR I...
 - Content-Type: text/html; charset=ISO-8859-1\r\n
 - Content-Encoding: gzip\r\n
 - Server: CAFE/1.0\r\n
 - Cache-control: private, x-gzip-ok=""\r\n
 - Content-length: 1272\r\n
 - Date: Thu, 13 May 2004 10:17:14 GMT\r\n
 - \r\n
 - [Request in frame: 18]
 - [Time since request: 0.971397000 seconds]
 - [Request URI [...]: /pagead/ads?client=ca-pub-2309191948673629&random=1084443430285&mt=1082467020&format=46...
 - [Full request URI [...]: http://pagead2.googlesyndication.com/pagead/ads?client=ca-pub-2309191948673629&rand...
 - Content-encoded entity body (gzip): 1272 bytes -> 3608 bytes
 - File Data: 3608 bytes

Application Data: This extension of the 5th layer can show the application-specific data.

```
Line-based text data: text/html (3 lines)
<html><head><style><!--\n
[...] .ch{cursor:pointer;cursor:hand}a.ad:link { color: #000000 }a.ad:visited { color: #000000 }a.a...
[...] function ss(w,id) {window.status = w;return true;}function cs(){window.status='';}function ca...
```

Answer the questions below

Use the "Exercise.pcapng" file to answer the questions. View packet number 38. Which markup language is used under the HTTP protocol?

eXtensible Markup Language

```
[TCP Flags: .....AP...]
Window size value: 6432
[Calculated window size: 6432]
[Window size scaling factor: -2 (no window scaling us
Checksum: 0x3d97 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
  [SEQ/ACK analysis]
  [Timestamps]
  TCP payload (424 bytes)
  TCP segment data (424 bytes)
  [14 Reassembled TCP Segments (18364 bytes): #6(1380), #
  Hypertext Transfer Protocol
  eXtensible Markup Language
    <?xml
    <!DOCTYPE html\n PUBLIC "-//W3C//DTD XHTML 1.0 Stric
    <html
```

What is the arrival date of the packet? 05/13/2004

```
Frame 38: 478 bytes on wire (3824 bits), 478 bytes capt
  Interface id: 0 (unknown)
  Encapsulation type: Ethernet (1)
  Arrival Time: May 13, 2004 10:17:12.158193000 UTC
  [Time shift for this packet: 0.000000000 seconds]
  Epoch Time: 1084443432.158193000 seconds
  [Time delta from previous captured frame: 0.070101000
  [Time delta from previous displayed frame: 0.07010100
  [Time since reference or first frame: 4.846969000 sec
```

What is the TTL value? 47

```

  ▸ Frame 38: 478 bytes on wire (3824 bits), 478 bytes captured
  ▸ Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00)
  ▾ Internet Protocol Version 4, Src: 65.208.228.223, Dst: 145.254.160.237
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 464
      Identification: 0xc0ac (49324)
    ▸ Flags: 0x4000, Don't fragment
      Fragment offset: 0
      Time to live: 47
      Protocol: TCP (6)
      Header checksum: 0x2fe0 [validation disabled]
      [Header checksum status: Unverified]
      Source: 65.208.228.223
      Destination: 145.254.160.237
    ▸ [Source GeoIP: US, ASN 17338, UNITAS-AOS]

```

What is the TCP payload size? 424

```

  ▾ Transmission Control Protocol, Src Port: 80, Dst Port: 3372
    Source Port: 80
    Destination Port: 3372
    [Stream index: 0]
    [TCP Segment Len: 424]
    Sequence number: 17941 (relative sequence number)
    Sequence number (raw): 290236320
    [Next sequence number: 18365 (relative sequence number)]
    Acknowledgment number: 480 (relative ack number)
    Acknowledgment number (raw): 951058419

```

What is the e-tag value?

9a01a-4696-7e354b00 (An ETag (entity tag) is an HTTP response header used to uniquely identify a specific version of a resource.)

```
Urgent pointer: 0
  ▶ [SEQ/ACK analysis]
  ▶ [Timestamps]
    TCP payload (424 bytes)
    TCP segment data (424 bytes)
  ▶ [14 Reassembled TCP Segments (18364 bytes): #6(1380), #1
  - Hypertext Transfer Protocol
    ▶ HTTP/1.1 200 OK\r\n
      Date: Thu, 13 May 2004 10:17:12 GMT\r\n
      Server: Apache\r\n
      Last-Modified: Tue, 20 Apr 2004 13:17:00 GMT\r\n
      ETag: "9a01a-4696-7e354b00"\r\n
      Accept-Ranges: bytes\r\n
    ▶ Content-Length: 18070\r\n
      Keep-Alive: timeout=15, max=100\r\n
      Connection: Keep-Alive\r\n
      Content-Type: text/html; charset=ISO-8859-1\r\n
```

Task 4 - Packet Navigation

Packet Numbers: Assign unique IDs to packets

http1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Start Stop Restart Options Open Save Close Reload Find Packet... Previous Packet Next Packet Go to Packet... First Packet

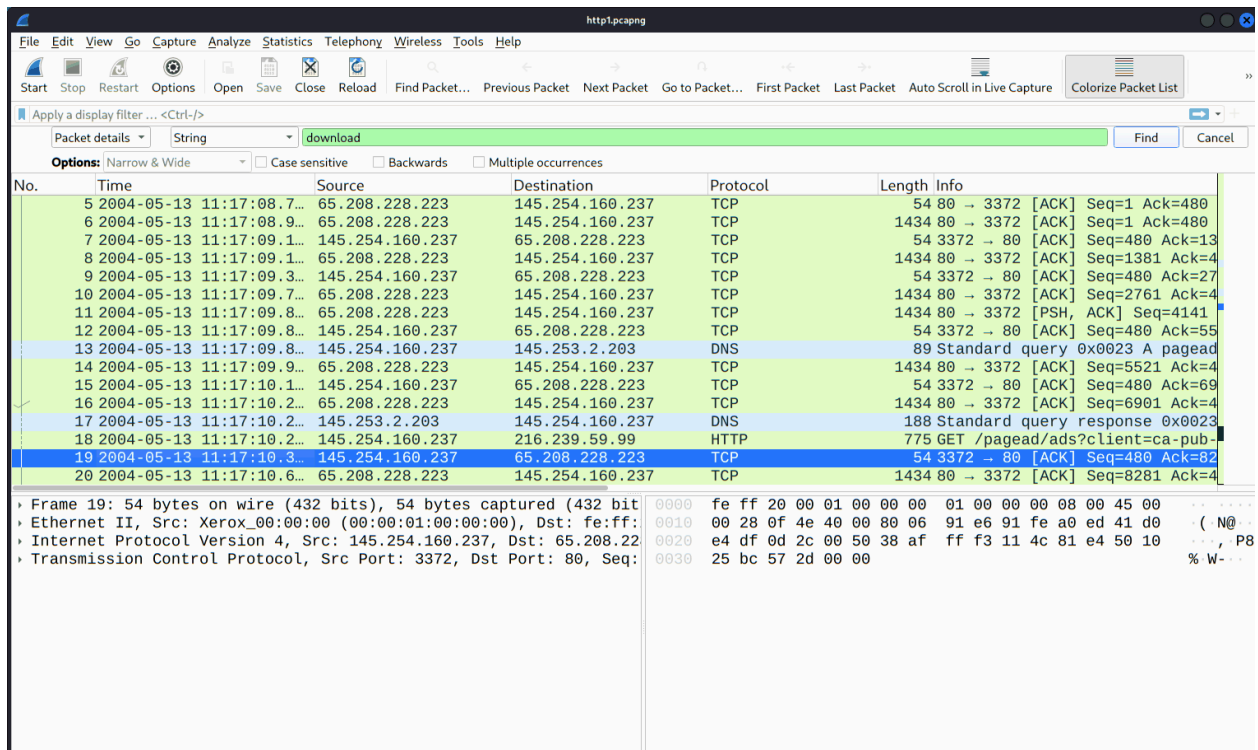
Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol
10	2004-05-13 11:17:09.7...	65.208.228.223	145.254.160.237	TCP
11	2004-05-13 11:17:09.8...	65.208.228.223	145.254.160.237	TCP
12	2004-05-13 11:17:09.8...	145.254.160.237	65.208.228.223	TCP
13	2004-05-13 11:17:09.8...	145.254.160.237	145.253.2.203	DNS
14	2004-05-13 11:17:09.9...	65.208.228.223	145.254.160.237	TCP
15	2004-05-13 11:17:10.1...	145.254.160.237	65.208.228.223	TCP
16	2004-05-13 11:17:10.2...	65.208.228.223	145.254.160.237	TCP
17	2004-05-13 11:17:10.2...	145.253.2.203	145.254.160.237	DNS
18	2004-05-13 11:17:10.2...	145.254.160.237	216.239.59.99	HTTP
19	2004-05-13 11:17:10.3...	145.254.160.237	65.208.228.223	TCP
20	2004-05-13 11:17:10.6...	65.208.228.223	145.254.160.237	TCP
21	2004-05-13 11:17:10.8...	65.208.228.223	145.254.160.237	TCP
22	2004-05-13 11:17:10.8...	145.254.160.237	65.208.228.223	TCP
23	2004-05-13 11:17:10.9...	65.208.228.223	145.254.160.237	TCP
24	2004-05-13 11:17:10.9...	216.239.59.99	145.254.160.237	TCP
25	2004-05-13 11:17:11.1...	145.254.160.237	65.208.228.223	TCP
26	2004-05-13 11:17:11.2...	216.239.59.99	145.254.160.237	TCP
27	2004-05-13 11:17:11.2...	216.239.59.99	145.254.160.237	HTTP
28	2004-05-13 11:17:11.2...	145.254.160.237	216.239.59.99	TCP
29	2004-05-13 11:17:11.4...	65.208.228.223	145.254.160.237	TCP
30	2004-05-13 11:17:11.5...	145.254.160.237	65.208.228.223	TCP

Frame 27: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface
 Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00), Dst: Xerox_00:00:00 (00:00:00:00:00:00)
 Internet Protocol Version 4, Src: 216.239.59.99, Dst: 145.254.160.237
 Transmission Control Protocol, Src Port: 80, Dst Port: 3371, Seq: 1431, Ack: 722, Len: 1590
 [2 Reassembled TCP Segments (1590 bytes): #26(1430), #27(160)]
 Hypertext Transfer Protocol
 Line-based text data: text/html (3 lines)

Go to Packet: Navigate directly to specific packets

Mark Packets: Flag packets for further analysis. **"Edit"** or the **"right-click"** menu to mark



Packet Comments: Annotate packets with investigation notes

Wireshark packet capture window showing a list of network packets. The selected packet (No. 12) is an Ethernet II frame, an Internet Protocol Version 4 packet, and a Transmission Control Protocol (TCP) segment. The packet list shows various protocols including TCP, DNS, and HTTP.

No.	Time	Source	Destination	Protocol	Length	Info
5	2004-05-13 11:17:08.7...	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK] Seq=1 Ack=480
6	2004-05-13 11:17:08.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1 Ack=480
7	2004-05-13 11:17:09.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=13
8	2004-05-13 11:17:09.1...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1381 Ack=4
9	2004-05-13 11:17:09.3...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=27
10	2004-05-13 11:17:09.7...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=2761 Ack=4
11	2004-05-13 11:17:09.8...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH, ACK] Seq=4141
12	2004-05-13 11:17:09.8...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=55
13	2004-05-13 11:17:09.8...	145.254.160.237	145.253.2.203	DNS	89	Standard query 0x0023 A pagead
14	2004-05-13 11:17:09.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=5521 Ack=4
15	2004-05-13 11:17:10.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=69
16	2004-05-13 11:17:10.2...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=6901 Ack=4
17	2004-05-13 11:17:10.2...	145.253.2.203	145.254.160.237	DNS	188	Standard query response 0x0023
18	2004-05-13 11:17:10.2...	145.254.160.237	216.239.59.99	HTTP	775	GET /pagead/ads?client=ca-pub-
19	2004-05-13 11:17:10.3...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=82
20	2004-05-13 11:17:10.6...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=8281 Ack=4
21	2004-05-13 11:17:10.8...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH, ACK] Seq=9661

Packet comments:

- Frame 12: 54 bytes on wire (432 bits), 54 bytes captured (432 bit)
- Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst: fe:ff:
- Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.22
- Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq:

Export Packets: Export selected packets for analysis

Wireshark packet capture window showing a list of network packets. The selected packet (No. 12) is an Ethernet II frame, an Internet Protocol Version 4 packet, and a Transmission Control Protocol (TCP) segment. The packet list shows various protocols including TCP, DNS, and HTTP.

No.	Time	Source	Destination	Protocol	Length	Info
1	2004-05-13 11:17:07.3...	145.254.160.237	65.208.228.223	TCP	62	3372 → 80 [SYN] Seq=0 Win=8760
2	2004-05-13 11:17:08.2...	65.208.228.223	145.254.160.237	TCP	62	80 → 3372 [SYN, ACK] Seq=0 Ack
3	2004-05-13 11:17:08.2...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=1 Ack=1 Wi
4	2004-05-13 11:17:08.2...	145.254.160.237	65.208.228.223	HTTP	533	GET /download.html HTTP/1.1
5	2004-05-13 11:17:08.7...	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK] Seq=1 Ack=480
6	2004-05-13 11:17:08.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1 Ack=480
7	2004-05-13 11:17:09.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=13
8	2004-05-13 11:17:09.1...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1381 Ack=4
9	2004-05-13 11:17:09.3...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=27
10	2004-05-13 11:17:09.7...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=2761 Ack=4
11	2004-05-13 11:17:09.8...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH, ACK] Seq=4141
12	2004-05-13 11:17:09.8...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=55
13	2004-05-13 11:17:09.8...	145.254.160.237	145.253.2.203	DNS	89	Standard query 0x0023 A pagead
14	2004-05-13 11:17:09.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=5521 Ack=4
15	2004-05-13 11:17:10.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=69
16	2004-05-13 11:17:10.2...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=6901 Ack=4
17	2004-05-13 11:17:10.2...	145.253.2.203	145.254.160.237	DNS	188	Standard query response 0x0023

Packet comments:

- Frame 12: 54 bytes on wire (432 bits), 54 bytes captured (432 bit)
- Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst: fe:ff:
- Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.22
- Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq:

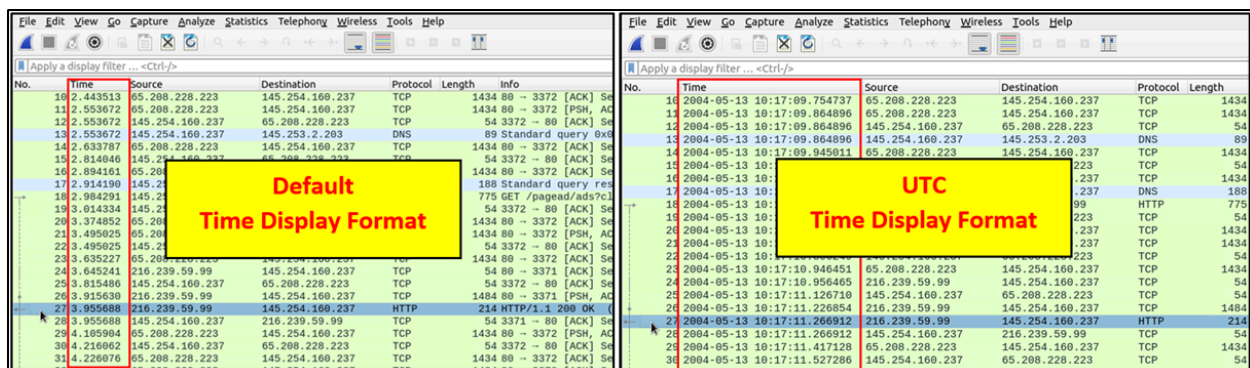
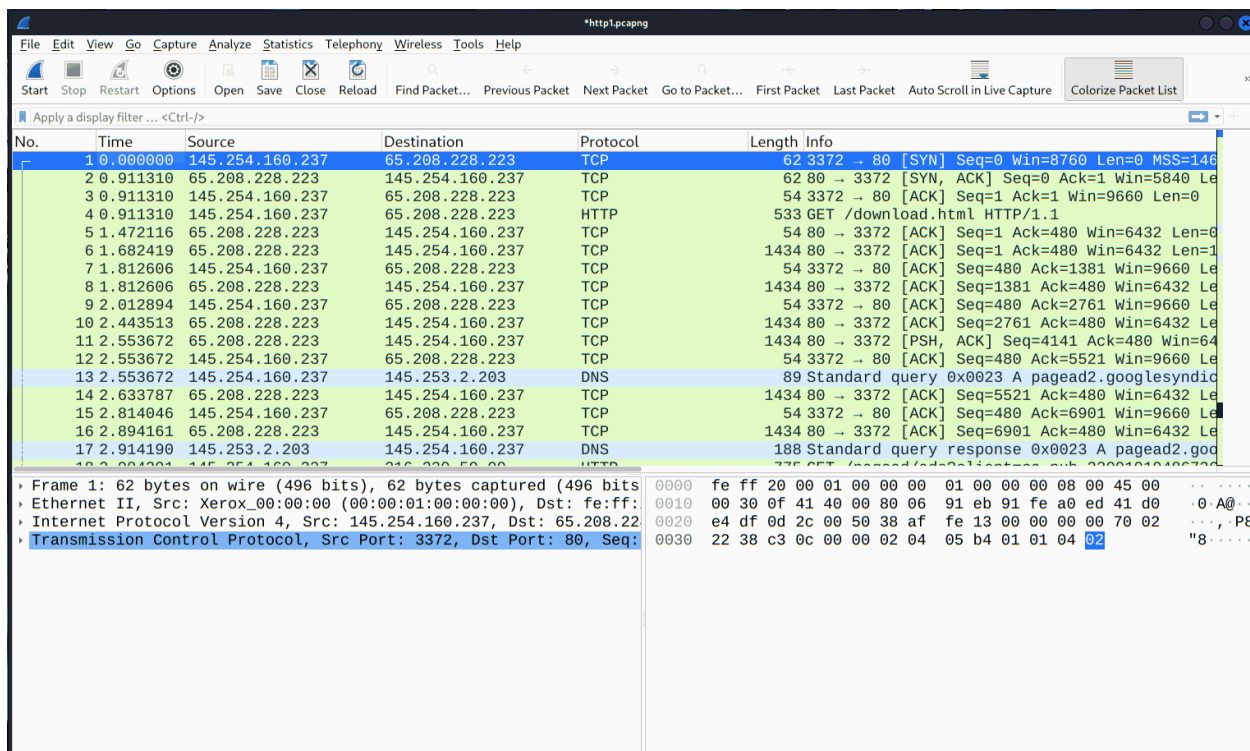
Export Objects (Files): Extract transferred files from network traffic

The screenshot shows the Wireshark interface with a packet capture of an HTTP GET request. The packet list shows a GET request for /download.html. The packet details pane shows the structure of the frame: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw hex and ASCII data.

No.	Time	Source	Destination	Protocol	Length	Info
1	2004-05-13 11:17:07.3...	145.254.160.237	65.208.228.223	TCP	62	3372 → 80 [SYN] Seq=0 Win=8760
2	2004-05-13 11:17:08.2...	65.208.228.223	145.254.160.237	TCP	62	80 → 3372 [SYN, ACK] Seq=0 Ack=
3	2004-05-13 11:17:08.2...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=1 Ack=1 Wi
4	2004-05-13 11:17:08.2...	145.254.160.237	65.208.228.223	HTTP	533	GET /download.html HTTP/1.1
5	2004-05-13 11:17:08.7...	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK] Seq=1 Ack=480
6	2004-05-13 11:17:08.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1 Ack=480
7	2004-05-13 11:17:09.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=13
8	2004-05-13 11:17:09.1...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=1381 Ack=4
9	2004-05-13 11:17:09.3...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=27
10	2004-05-13 11:17:09.7...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=2761 Ack=4
11	2004-05-13 11:17:09.8...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH, ACK] Seq=4141
12	2004-05-13 11:17:09.8...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=55
13	2004-05-13 11:17:09.8...	145.254.160.237	145.253.2.203	DNS	89	Standard query 0x0023 A pagead
14	2004-05-13 11:17:09.9...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=5521 Ack=4
15	2004-05-13 11:17:10.1...	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=69
16	2004-05-13 11:17:10.2...	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=6901 Ack=4
17	2004-05-13 11:17:10.2...	145.253.2.203	145.254.160.237	DNS	188	Standard query response 0x0023

Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface 0
Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst: fe:ff:fe:00:00:00
Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.228.223
Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq: 0

Time Display Format: Display timestamps in different formats. View → Time Display Format



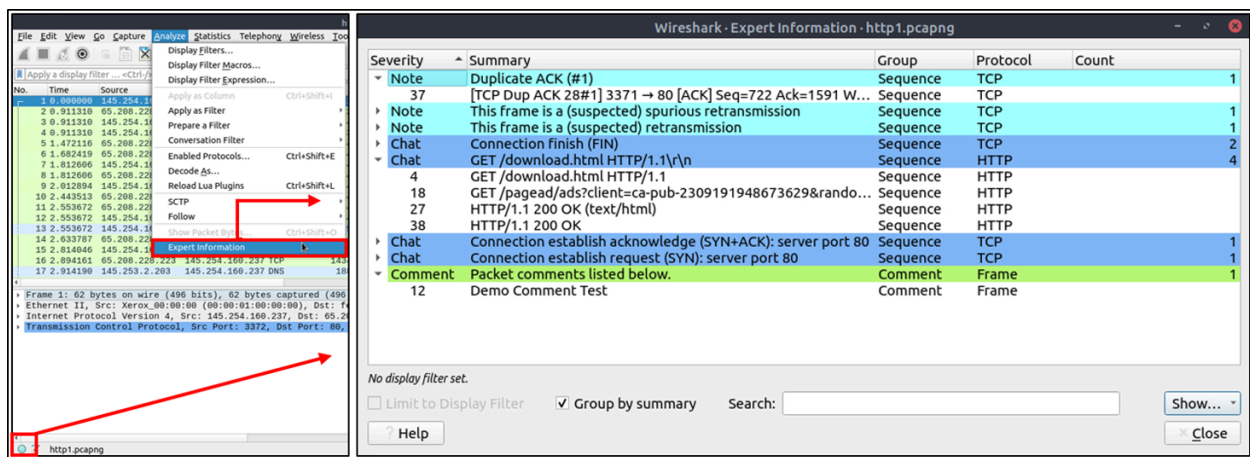
Expert Info: Identify protocol errors and anomalies

Severity	Colour	Info
Chat	Blue	Information on usual workflow.
Note	Cyan	Notable events like application error codes.
Warn	Yellow	Warnings like unusual error codes or problem statements.
Error	Red	Problems like malformed packets

Group	Info	Group	Info
-------	------	-------	------

Checksum	Checksum errors	Deprecated	Deprecated protocol usage
Comment	Packet comment detection	Malformed	Malformed packet detection

You can use the **"lower left bottom section"** in the status bar or **"Analyze → Expert Information"** menu to view all available information entries via a dialogue box. It will show the packet number, summary, group protocol and total occurrence.



Answer the questions below

Use the "Exercise.pcapng" file to answer the questions. Search the "r4w" string in packet details. What is the name of artist 1?

r4w8173

current filter: tcp.stream eq 12

Packet details	Narrow & Wide	Case sensitive	String	r4w	Find	Cancel
Time	Source	Destination	Protocol	Length	Info	
31192	568415496.92...	44.228.249.3	10.10.57.178	HTTP	2625	HTTP/1.1 200 OK (
31193	568415496.92...	10.10.57.178	44.228.249.3	TCP	66	57672 → 80 [ACK] S
32609	568415498.98...	10.10.57.178	44.228.249.3	HTTP	469	GET /categories.ph
32630	568415499.11...	44.228.249.3	10.10.57.178	TCP	66	80 → 57672 [ACK] S
32631	568415499.11...	44.228.249.3	10.10.57.178	HTTP	2813	HTTP/1.1 200 OK (
32632	568415499.11...	10.10.57.178	44.228.249.3	TCP	66	57672 → 80 [ACK] S
33744	568415501.16...	10.10.57.178	44.228.249.3	HTTP	480	GET /artists.php H
33789	568415501.28...	44.228.249.3	10.10.57.178	TCP	66	80 → 57672 [ACK] S
33790	568415501.28...	44.228.249.3	10.10.57.178	HTTP	2670	HTTP/1.1 200 OK (
33791	568415501.28...	10.10.57.178	44.228.249.3	TCP	66	57672 → 80 [ACK] S
35437	568415504.78...	10.10.57.178	44.228.249.3	HTTP	480	GET /categories.ph
35531	568415504.90...	44.228.249.3	10.10.57.178	TCP	66	80 → 57672 [ACK] S
35532	568415504.91...	44.228.249.3	10.10.57.178	HTTP	2813	HTTP/1.1 200 OK (
35533	568415504.91...	10.10.57.178	44.228.249.3	TCP	66	57672 → 80 [ACK] S
37036	568415507.39...	10.10.57.178	44.228.249.3	HTTP	491	GET /listproducts.
37060	568415507.51...	44.228.249.3	10.10.57.178	TCP	2962	80 → 57672 [PSH, A
37061	568415507.51...	44.228.249.3	10.10.57.178	HTTP	143	HTTP/1.1 200 OK (
37062	568415507.51...	10.10.57.178	44.228.249.3	TCP	66	57672 → 80 [ACK] S

0930 69 65 2e 0a 53 65 64 20 61 6c 69 71 70
 0940 73 65 6d 20 75 74 20 61 72 63 75 2e 3
 0950 3c 70 3e 70 61 69 6e 74 65 64 20 62 7
 0960 61 20 68 72 65 66 3d 27 61 72 74 69 7
 0970 70 68 70 3f 61 62 74 69 73 74 3d 31 2
 0980 77 38 31 37 33 3c 2f 61 3e 3c 2f 70 3
 0990 3c 61 20 68 72 65 66 3d 27 23 27 20 6

```
ubuntu@ip-10-112-169-244:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
ubuntu@ip-10-112-169-244:~$ cd Desktop
ubuntu@ip-10-112-169-244:~/Desktop$ ls
2.jpg Exercise.pcapng http1.pcapng mate-terminal.desktop wireshark.desktop
ubuntu@ip-10-112-169-244:~/Desktop$ sha256sum 2.jpg
99958e145afe69a59b03f53712e6084b788ec09bed4ad9094a941c0c438942a3 2.jpg
ubuntu@ip-10-112-169-244:~/Desktop$ md5sum 2.jpg
911cd574a42865a956ccde2d04495ebf 2.jpg
ubuntu@ip-10-112-169-244:~/Desktop$ S
```

Look at the expert info section. What is the number of warnings? 1636

The image shows the Wireshark interface with the Expert Information window open. The window title is "Wireshark · Expert Information · Exercise.pcapng". It displays a list of messages with columns for Severity, Summary, and Group. The messages are as follows:

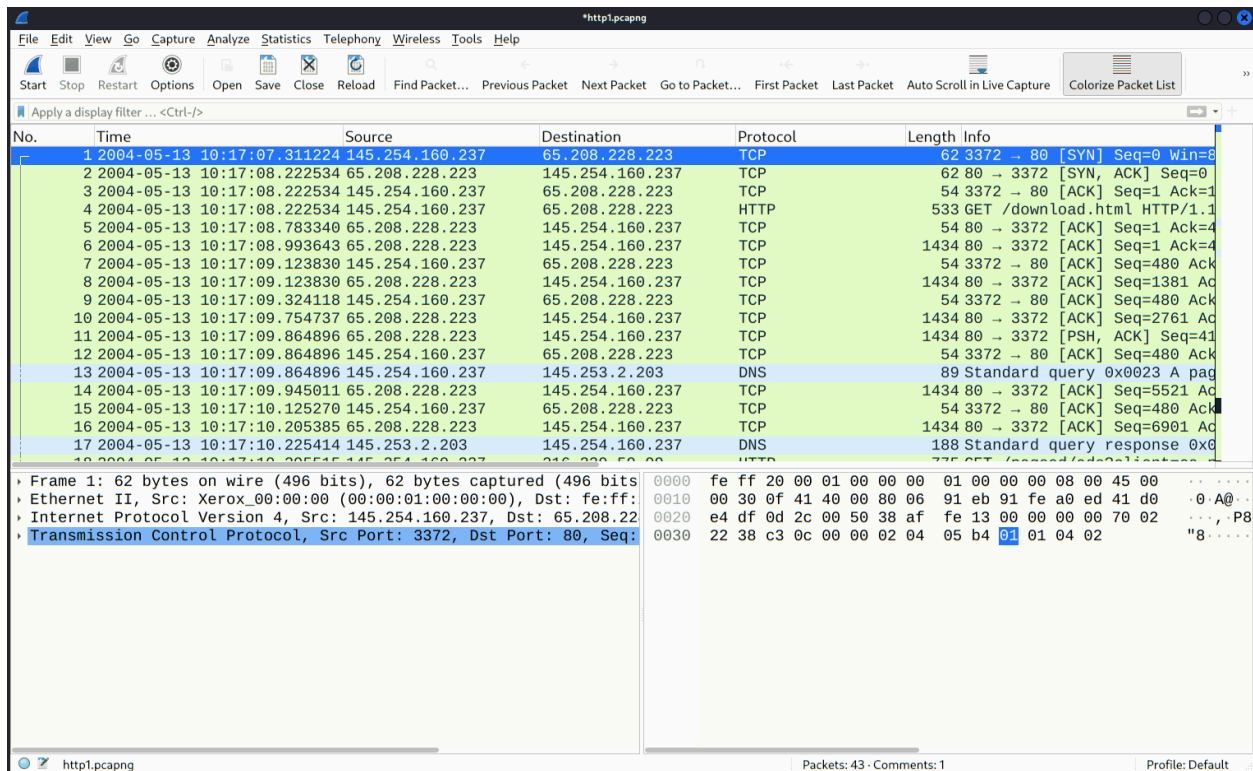
Severity	Summary	Group
Error	Malformed Packet (Exception occurred)	Malforme
Error	Malformed Packet (Exception occurred)	Malforme
Warning	Illegal characters found in header name	Protocol
Note	ACK to a TCP keep-alive segment	Sequence
Note	TCP keep-alive segment	Sequence
Note	Duplicate ACK (#1)	Sequence
Note	This frame is a (suspected) spurious retransmission	Sequence
Note	This frame is a (suspected) retransmission	Sequence
Chat	Connection finish (FIN)	Sequence
Chat	GET /download.html HTTP/1.1\r\n	Sequence
Chat	Connection establish acknowledge (SYN+ACK): server port 80	Sequence
Chat	Connection establish request (SYN): server port 80	Sequence
Comment	Packet comments listed below.	Comment

The "Warning" message is highlighted in yellow. The background shows the packet list with packet 4267 selected, and the packet details pane showing the structure of the selected packet: Frame 4267: 5520 bytes, Ethernet II, Src: 08:00:27:00:00:00, Internet Protocol Version 4, Transmission Control Protocol, [2 Reassembled TCP Segments (40000-40001) → 40002], Hypertext Transfer Protocol, Line-based text data.

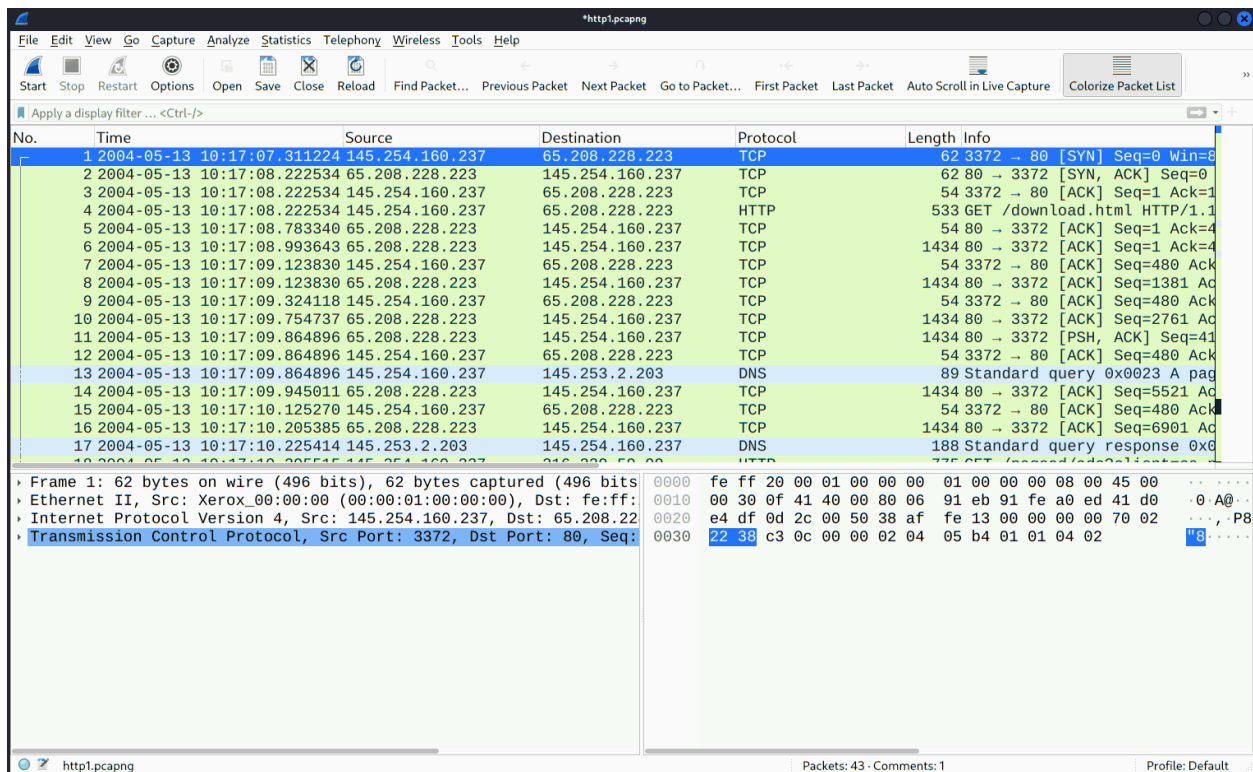
Task 5 - Packet Filtering

Packet Filtering: Filter traffic to reduce noise

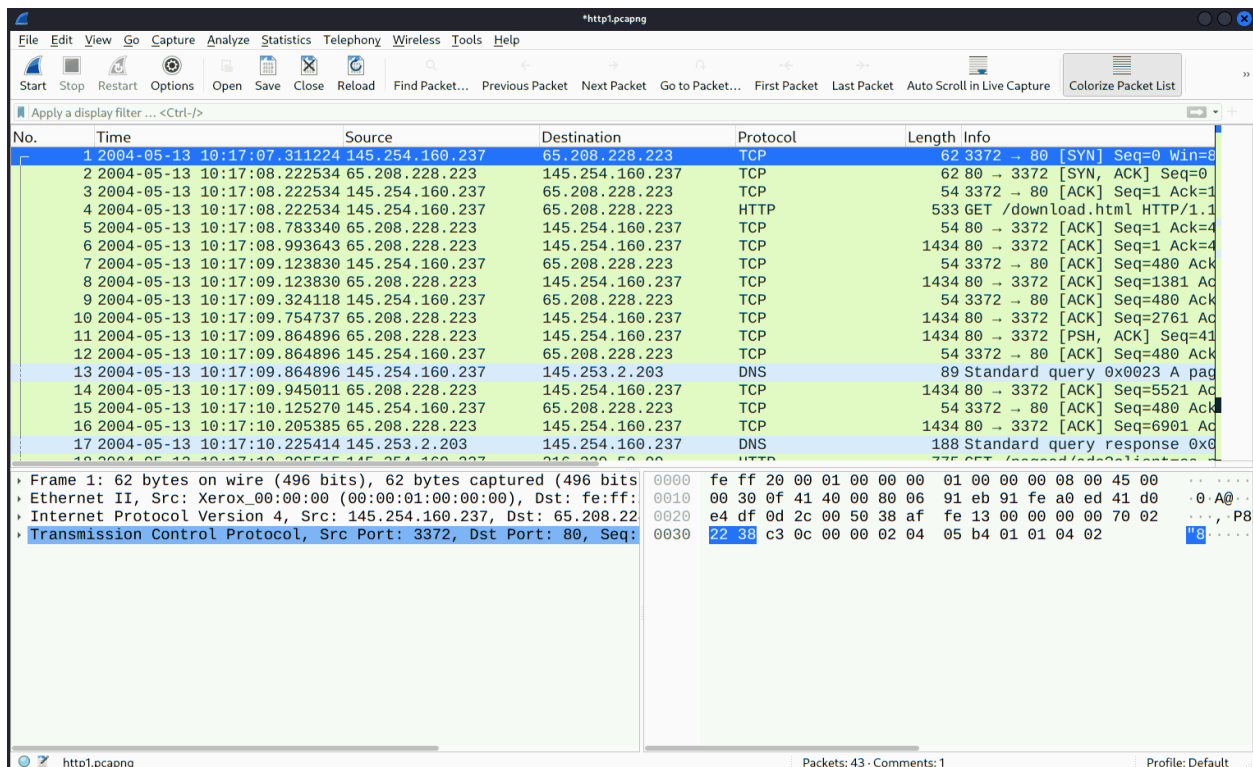
Apply as Filter: Use queries or right-click filters. **Analyse → Apply as Filter**



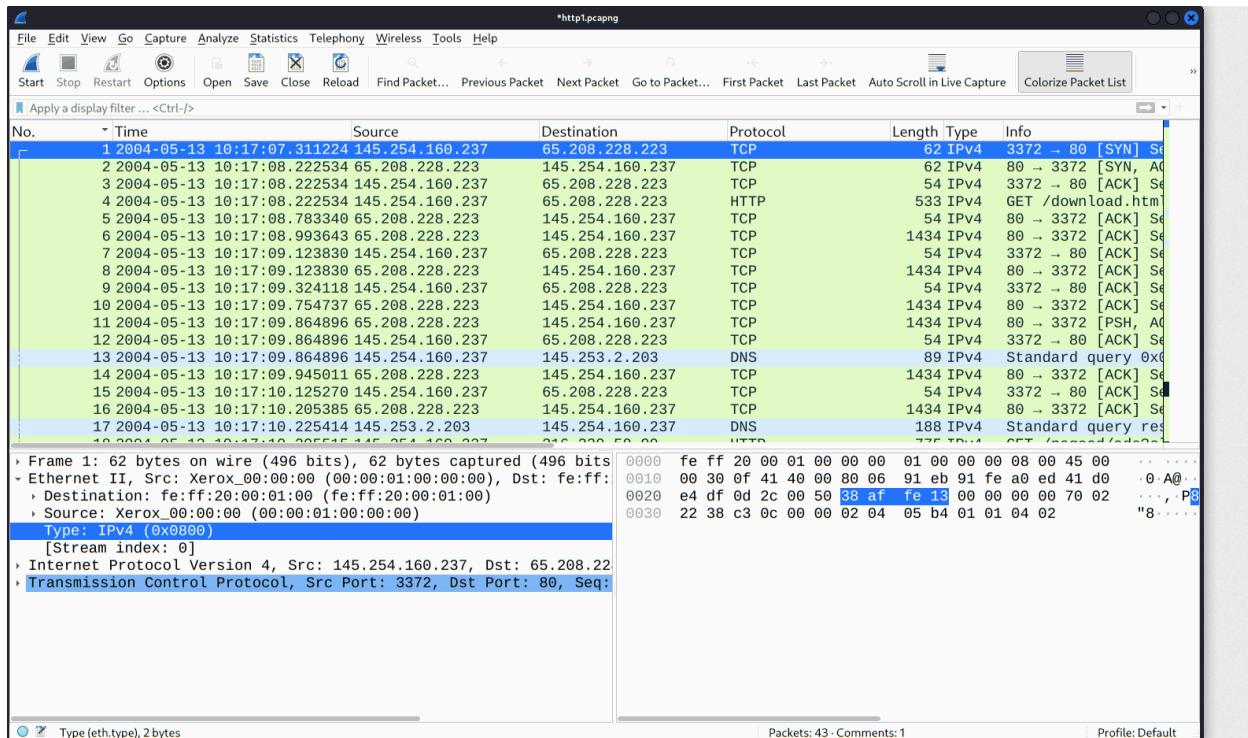
Conversation Filter: Show related conversation packets. **Analyse → Conversation Filter**



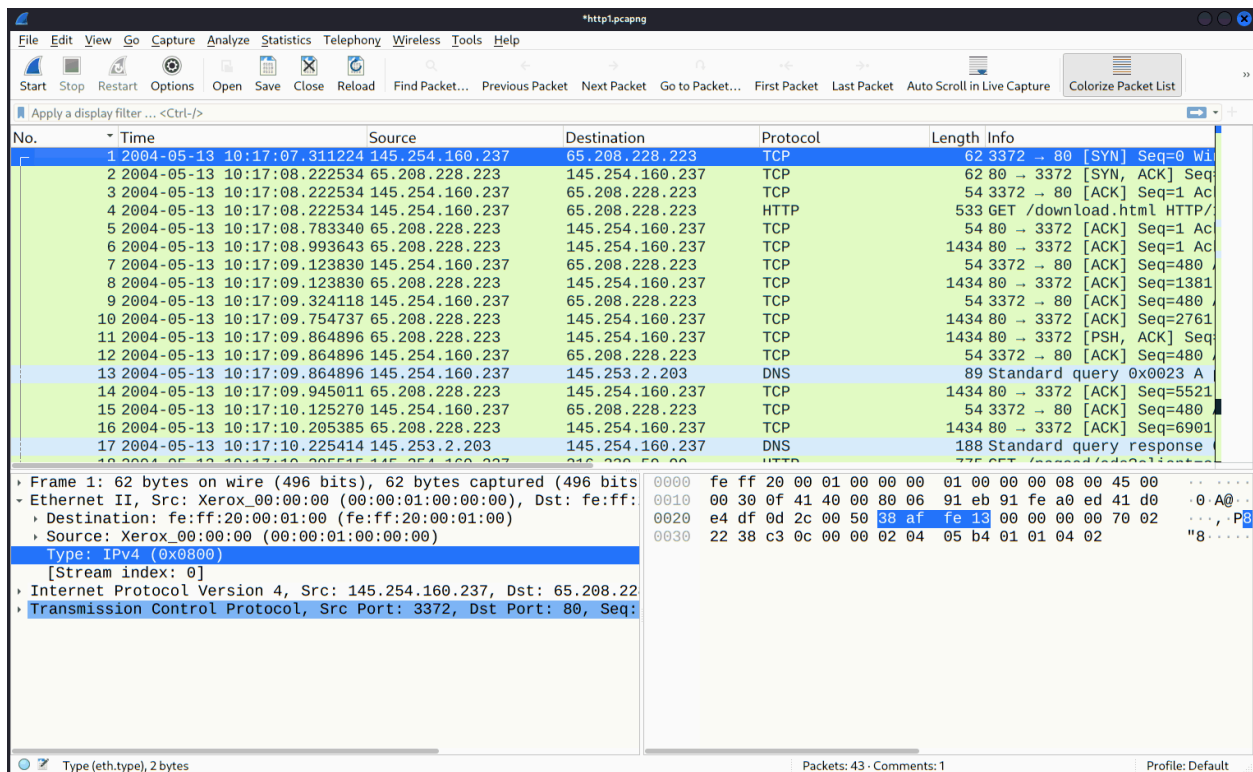
Colourise Conversation: Highlight conversation without filtering. **View → Colourise Conversation → Reset Colourisation**



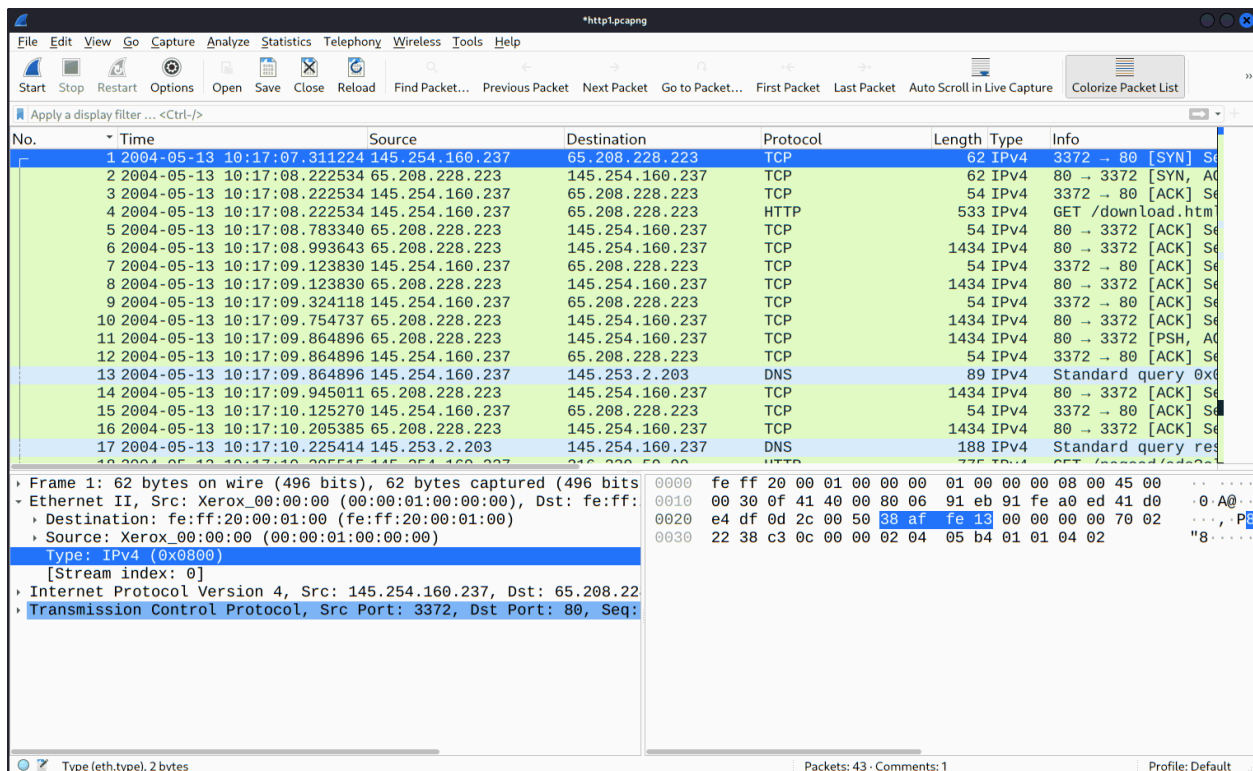
Prepare as Filter: Build filters before applying



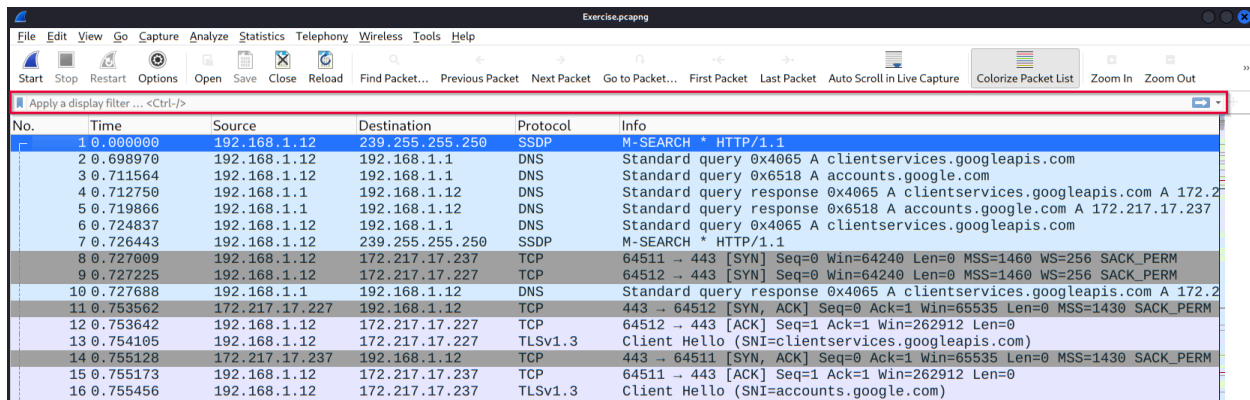
Apply as Column: Display selected fields as columns. **Analyse → Apply as Column**



Follow Stream: Reconstruct application-layer communication. Analyse → Follow TCP/UDP/HTTP Stream

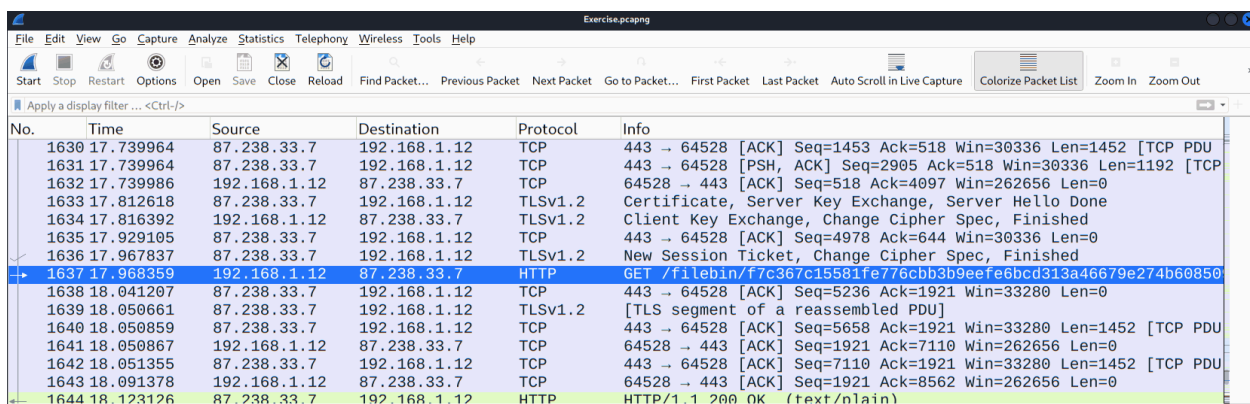


Simple Display Filter Queries: Quickly filter packet captures.



Filter By Protocol Name or Port: Filter traffic by protocol or port. There are two basic ways to filter based on a specific protocol: by protocol name and by protocol port number.

To filter by **protocol name**, simply type in the protocol name and hit enter or click on the arrow button at the right hand side of the display filter bar. The GIF below shows an example of how to filter for http traffic. You can similarly filter for other protocols using keywords like `arp`, `dhcp`, `ftp`, `smtp`, `pop`, `imap`, and more.



To filter by **protocol port number**, you can use the structure "`tcp.port == <port number>`" or "`udp.port == <port number>`". For example, if you want to see only http packets, you would use the filter "`tcp.port == 80`" and then hit enter.

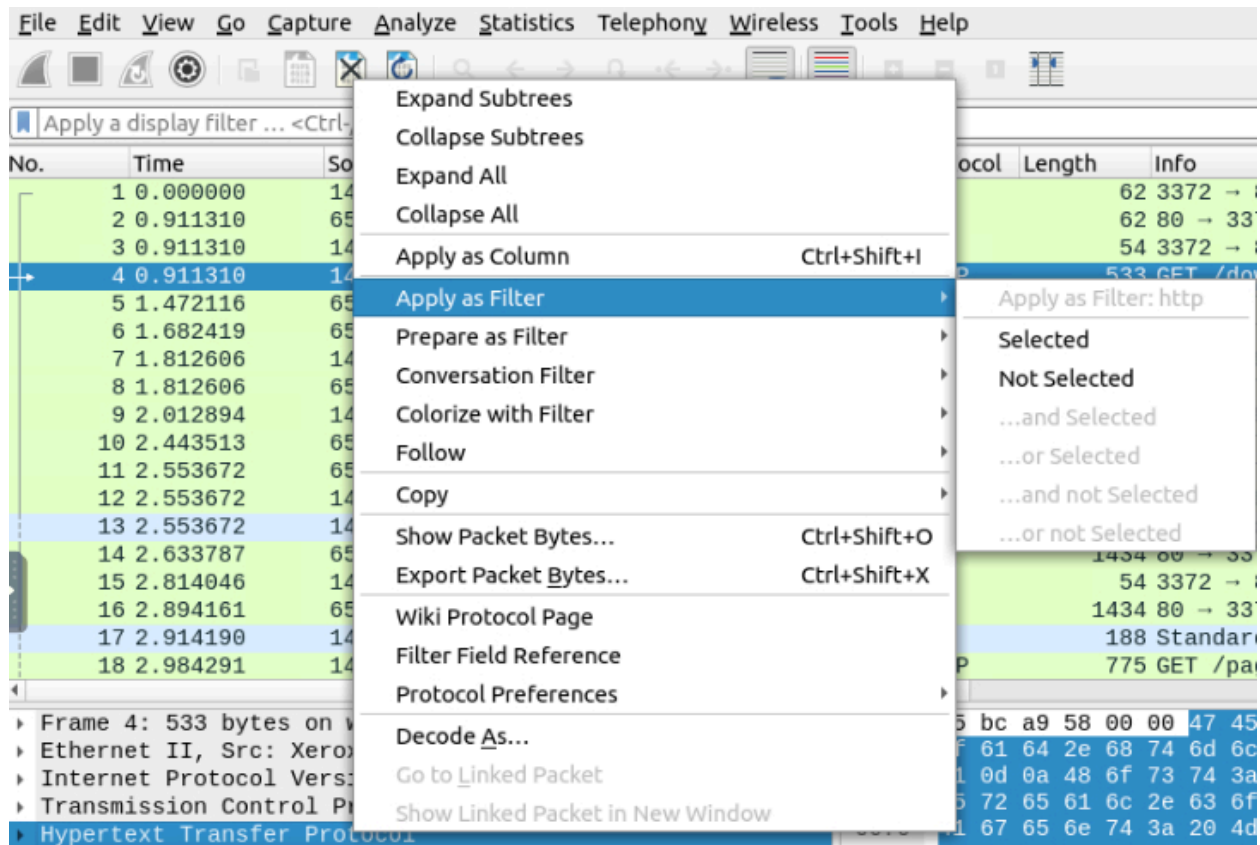
No.	Time	Source	Destination	Protocol	Info
43	478.695815	27.124.5.16	198.71.247.91	TCP	58204 → 2078 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44	485.287010	45.134.26.236	198.71.247.91	TCP	43963 → 30195 [SYN] Seq=0 Win=1024 Len=0
45	515.425164	45.146.164.101	198.71.247.91	TCP	51282 → 3408 [SYN] Seq=0 Win=1024 Len=0
46	528.855476	103.207.36.91	198.71.247.91	TCP	53443 → 3344 [SYN] Seq=0 Win=1024 Len=0
47	535.494953	43.131.66.209	198.71.247.91	TCP	16378 → 6240 [SYN] Seq=0 Win=65535 Len=0 MSS=1424 WS=256 SACK_P
48	545.777882	8.218.122.85	198.71.247.91	MDNS	Unknown operation (13) 0x542e[Malformed Packet]
49	560.686794	91.223.67.180	198.71.247.91	TCP	43728 → 22322 [SYN] Seq=0 Win=1024 Len=0
50	571.039212	141.101.105.7	198.71.247.91	TCP	9418 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM WS=102
51	571.039625	198.71.247.91	141.101.105.7	TCP	80 → 9418 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_
52	571.190014	141.101.105.7	198.71.247.91	TCP	9418 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
53	571.190073	141.101.105.7	198.71.247.91	HTTP	GET /.env HTTP/1.1
54	571.190317	198.71.247.91	141.101.105.7	TCP	80 → 9418 [ACK] Seq=1 Ack=482 Win=64128 Len=0
55	571.192601	198.71.247.91	141.101.105.7	HTTP	HTTP/1.1 404 Not Found (text/html)
56	571.342926	141.101.105.7	198.71.247.91	TCP	9418 → 80 [ACK] Seq=482 Ack=497 Win=67584 Len=0
57	571.495729	141.101.76.61	198.71.247.91	TCP	53702 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM WS=10

Filter By IP: Filter traffic by IP address.

No.	Time	Source	Destination	Protocol	Info
50	571.039212	141.101.105.7	198.71.247.91	TCP	9418 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM WS=102
51	571.039625	198.71.247.91	141.101.105.7	TCP	80 → 9418 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_
52	571.190014	141.101.105.7	198.71.247.91	TCP	9418 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
53	571.190073	141.101.105.7	198.71.247.91	HTTP	GET /.env HTTP/1.1
54	571.190317	198.71.247.91	141.101.105.7	TCP	80 → 9418 [ACK] Seq=1 Ack=482 Win=64128 Len=0
55	571.192601	198.71.247.91	141.101.105.7	HTTP	HTTP/1.1 404 Not Found (text/html)
56	571.342926	141.101.105.7	198.71.247.91	TCP	9418 → 80 [ACK] Seq=482 Ack=497 Win=67584 Len=0
57	571.495729	141.101.76.61	198.71.247.91	TCP	53702 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM WS=10
58	571.495797	198.71.247.91	141.101.76.61	TCP	80 → 53702 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_
59	571.670147	141.101.76.61	198.71.247.91	TCP	53702 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
60	571.670171	141.101.76.61	198.71.247.91	TCP	53702 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=547 [TCP PDU re
61	571.670216	198.71.247.91	141.101.76.61	TCP	80 → 53702 [ACK] Seq=1 Ack=548 Win=64128 Len=0
62	571.670248	141.101.76.61	198.71.247.91	HTTP	POST / HTTP/1.1 (application/x-www-form-urlencoded)
63	571.670254	198.71.247.91	141.101.76.61	TCP	80 → 53702 [ACK] Seq=1 Ack=568 Win=64128 Len=0
64	571.671117	198.71.247.91	141.101.76.61	HTTP	HTTP/1.1 200 OK (text/html)

Answer the questions below

Use the "Exercise.pcapng" file to answer the questions. Go to packet number 4. Right-click on the "Hypertext Transfer Protocol" and apply it as a filter. Now, look at the filter pane. What is the filter query? http



What is the number of displayed packets? 1089

http

No.	Time	Source	Destination	Protocol	Length
4	0.911310	145.254.160.237	65.208.228.223	HTTP	
18	2.984291	145.254.160.237	216.239.59.99	HTTP	
27	3.955688	216.239.59.99	145.254.160.237	HTTP	
38	4.846969	65.208.228.223	145.254.160.237	HTTP/X...	
643	568415438.43...	10.10.57.178	10.100.1.33	TCP	9
665	568415438.44...	10.10.57.178	10.100.1.33	TCP	9
676	568415438.44...	10.10.57.178	10.100.1.33	TCP	9
700	568415438.44...	10.10.57.178	10.100.1.33	TCP	9
768	568415438.63...	10.10.57.178	10.100.1.33	TCP	9
797	568415438.64...	10.10.57.178	10.100.1.33	TCP	9
821	568415438.80...	10.10.57.178	10.100.1.33	TCP	6
870	568415438.95...	10.10.57.178	10.100.1.33	TCP	9
871	568415438.95...	10.10.57.178	10.100.1.33	TCP	9
905	568415438.96...	10.10.57.178	10.100.1.33	TCP	9
950	568415439.10...	10.10.57.178	10.100.1.33	TCP	9
959	568415439.10...	10.10.57.178	10.100.1.33	TCP	9
971	568415439.11...	10.10.57.178	10.100.1.33	TCP	9
993	568415439.17...	10.10.57.178	10.100.1.33	TCP	2

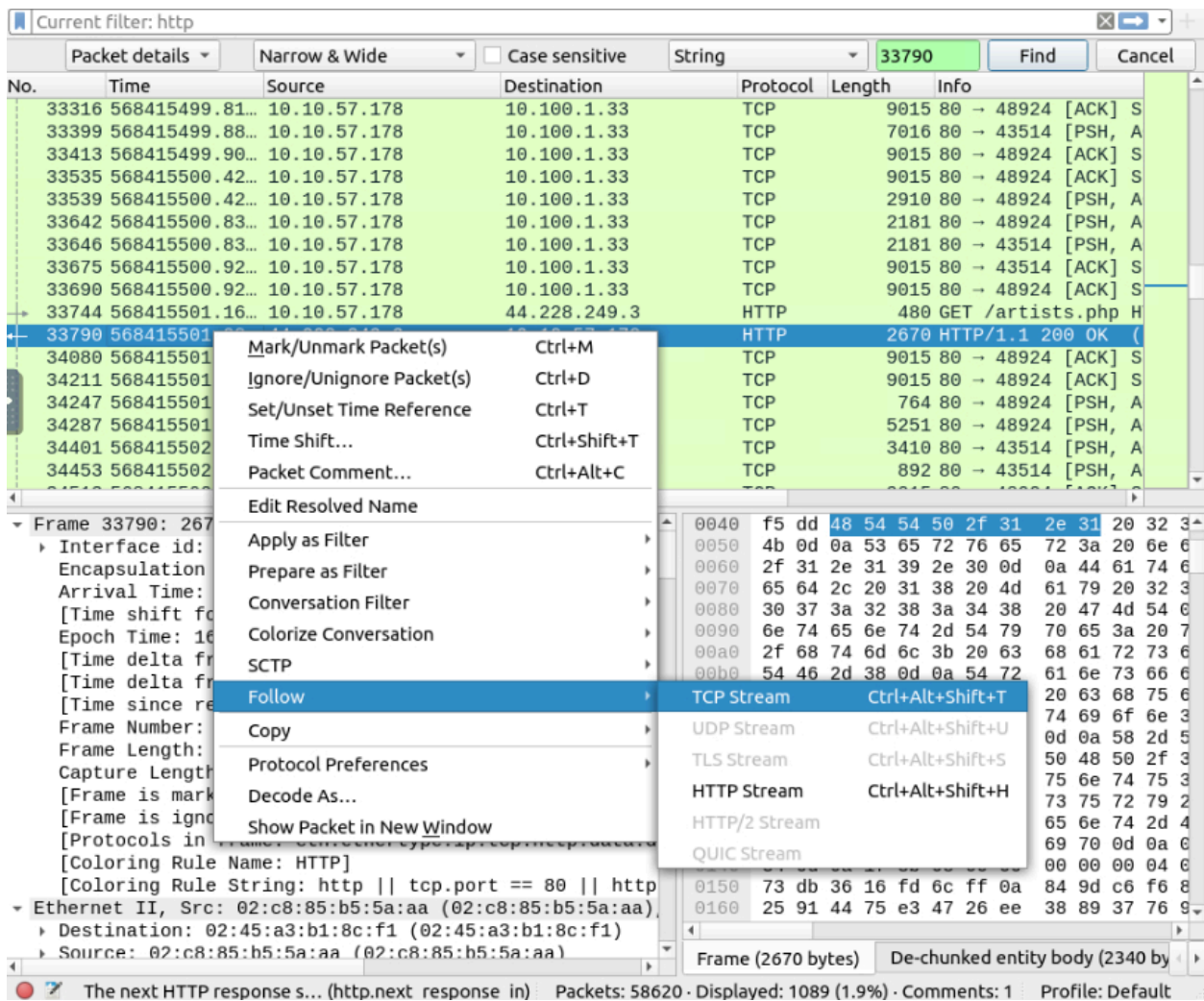
▶ Frame 4: 533 bytes on wire (4264 bits), 533 bytes capture
 ▶ Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst:
 ▶ Internet Protocol Version 4, Src: 145.254.160.237, Dst: 6
 ▶ Transmission Control Protocol, Src Port: 3372, Dst Port:
 ▶ Hypertext Transfer Protocol

0030	25 bc a9 58
0040	6f 61 64 2e
0050	31 0d 0a 48
0060	65 72 65 61
0070	41 67 65 6e
0080	2e 30 20 28
0090	57 69 6e 64
00a0	65 6e 2d 55
00b0	65 63 6b 6f
00c0	63 63 65 70
00d0	61 70 70 6c
00e0	61 70 70 6c
00f0	6c 2b 78 6d
0100	71 3d 30 2e
0110	3b 71 3d 30
0120	2c 69 6d 61
0130	65 2f 67 69
0140	71 3d 30 2e
0150	6e 67 75 61
0160	3b 71 3d 30
0170	6e 63 6f 64
0180	66 6c 61 7a

Hypertext Transfer Protocol (http), 479 byte(s) Packets: 58620 · Displayed: 1089 (1.9%)

Go to packet number 33790, follow the HTTP stream, and look carefully at the responses.

Looking at the web server's response, what is the total number of artists? 3



What is the name of the second artist? Blad3

